

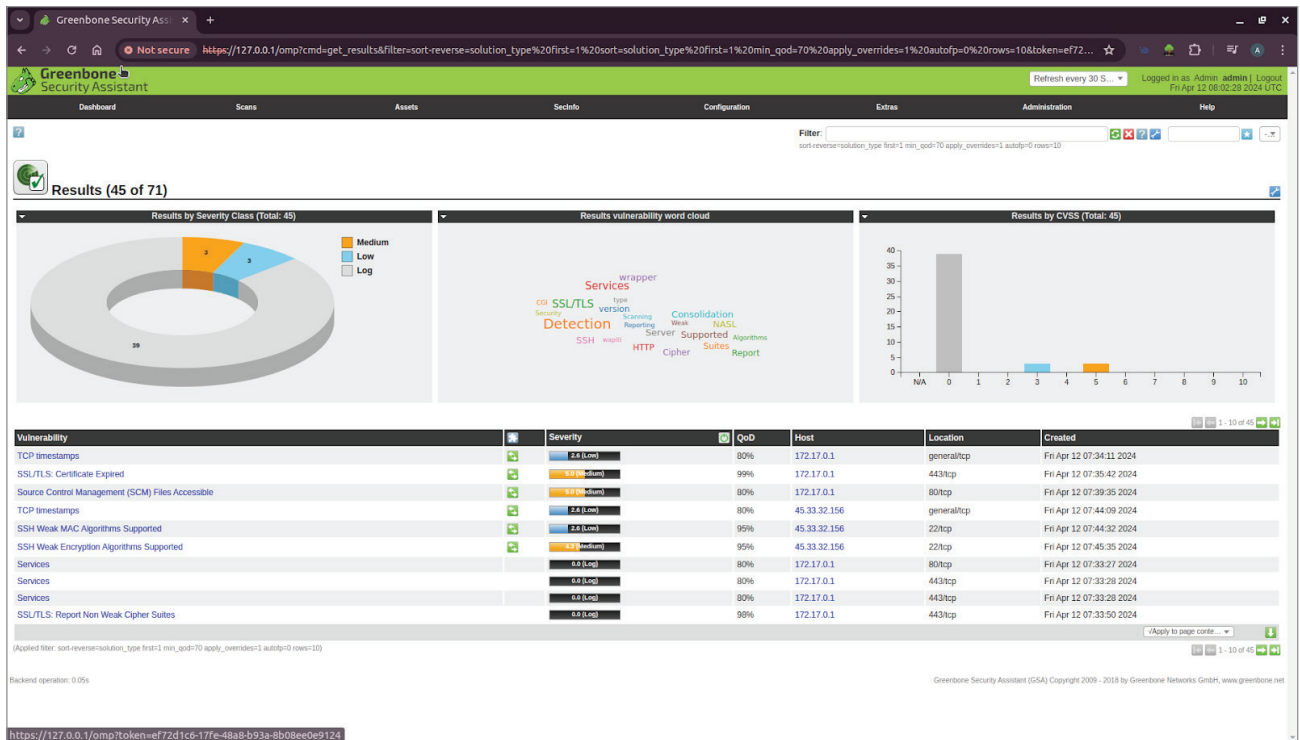


Figure 6: Prioritising vulnerabilities

Vulnerability	Severity
TCP timestamps	2.6 (Low)
SSL/TLS: Certificate Expired	5.0 (Medium)
Source Control Management (SCM) Files Accessible	5.0 (Medium)
TCP timestamps	2.6 (Low)
SSH Weak MAC Algorithms Supported	2.6 (Low)
SSH Weak Encryption Algorithms Supported	4.3 (Medium)

Figure 7: List of vulnerabilities and their severity

credentials, session data, and potentially confidential business information. Attackers could use this information to gain unauthorised access to systems or conduct further attacks within the network.

Mitigation strategies

TCP timestamps:

- Disable TCP timestamps if these are not required for specific applications or services.
- Implement network filtering to block external access to TCP timestamps where possible.
- Regularly update and patch systems to mitigate known vulnerabilities associated with TCP timestamps.

SSL/TLS certificate expired:

Implement a robust certificate management process to ensure certificates are renewed before expiration.

- Use monitoring tools to alert administrators about upcoming certificate expirations.
- Employ certificate transparency logs to detect unauthorised issuance of certificates for your domain.
- Utilise technologies such as HTTP Strict Transport Security (HSTS) to enforce secure connections and mitigate the impact of expired certificates.

SCM files accessible:

- Implement access controls and

permissions on SCM repositories to restrict access to authorised personnel only.

- Regularly audit and review access logs for SCM repositories to detect unauthorised access attempts.
- Encrypt sensitive information stored in SCM files to protect it from unauthorised disclosure.
- Utilise tools and services that automatically scan repositories for sensitive information and potential vulnerabilities.

SSH weak MAC algorithms supported:

- Disable weak MAC algorithms in SSH configurations and only allow the use of strong, cryptographically secure MAC algorithms.
- Regularly review and update SSH configurations to ensure they adhere to security best practices.
- Implement network segmentation and access controls to limit SSH access to authorised users and devices.
- Monitor SSH traffic for signs of malicious activity, such as unauthorised access attempts or